

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Strategic insights across demographics, purchases & behavior





Dataset at a Glance

3900

Transactions

18

Columns

4

Categories

Clothing, Footwear, Accessories,
Outerwear

Key Data Features

Demographics

Age, Gender, Location,
Subscription Status, Age Group

Purchase Details

Item, Category, Amount, Color,
Season

Shopping Behavior

Rating, Shipping, Discount,
Payment Method, Frequency

EDA in Python

● Load & Explore

used pandas → `df.info()`,
`df.describe()`, `df.head()`
`df.isnull().sum()`

● Standardize

Snake_case columns ·
renamed fields

● Feature Engineering

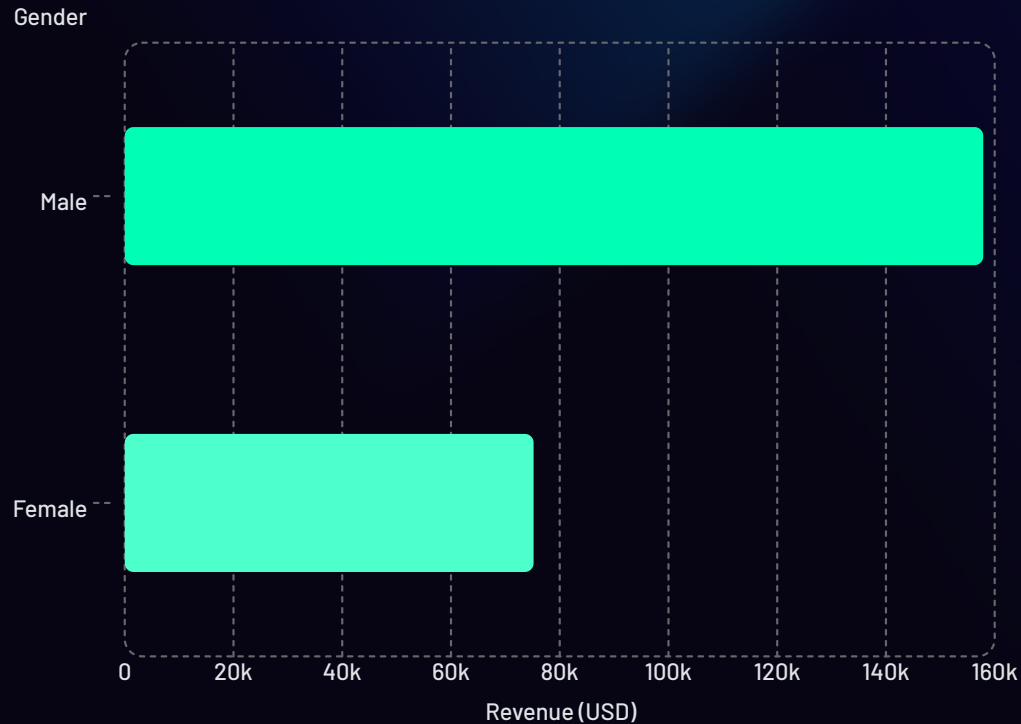
age_group bins ·
purchase_frequency_da
ys mapping

● Clean & Export

Dropped redundant
column -
promo_code_used →
CSV



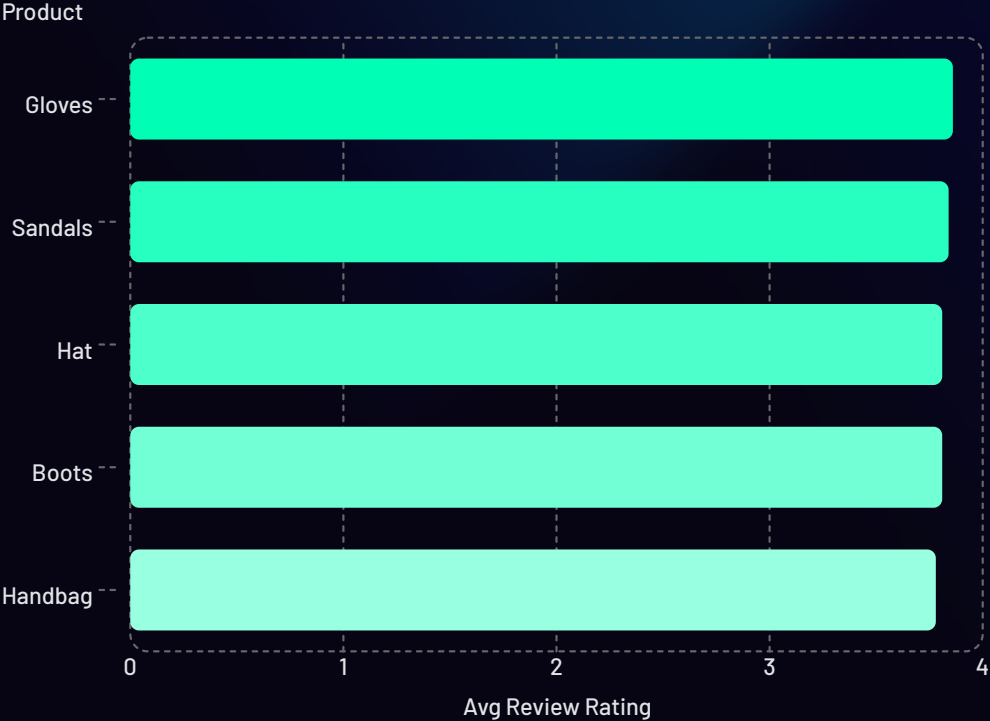
Revenue & High Spenders



Male customers generate 2× more revenue

high-spending discount users identified — purchases above average even with discounts applied.

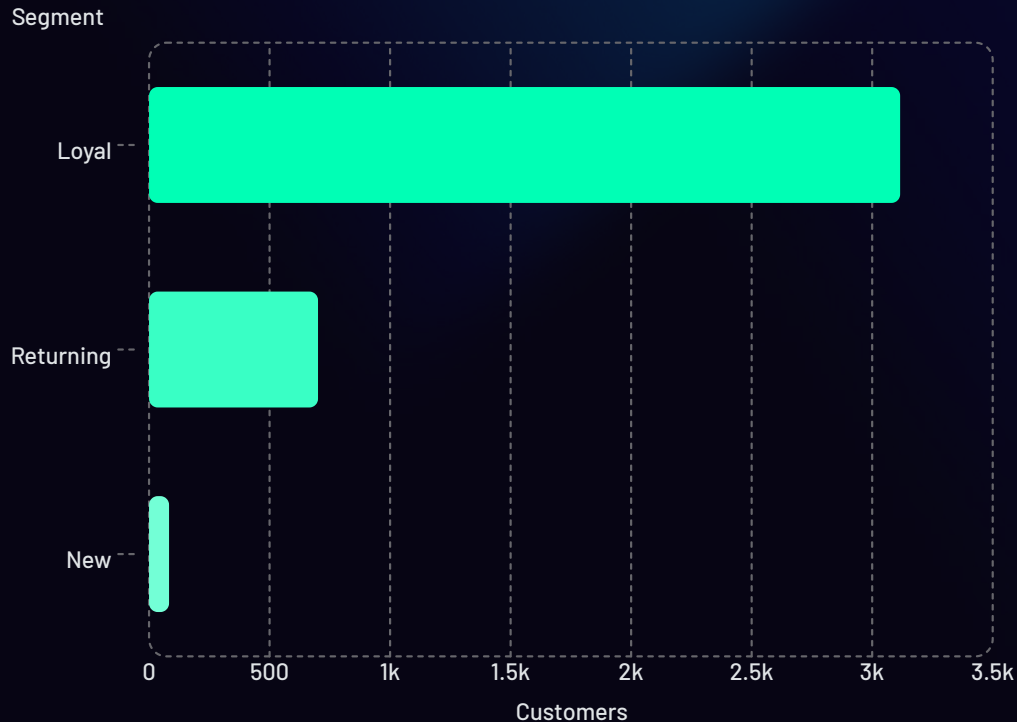
Top Products by Discount



Top Discounted Products

Item	Discount %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation & Shipping



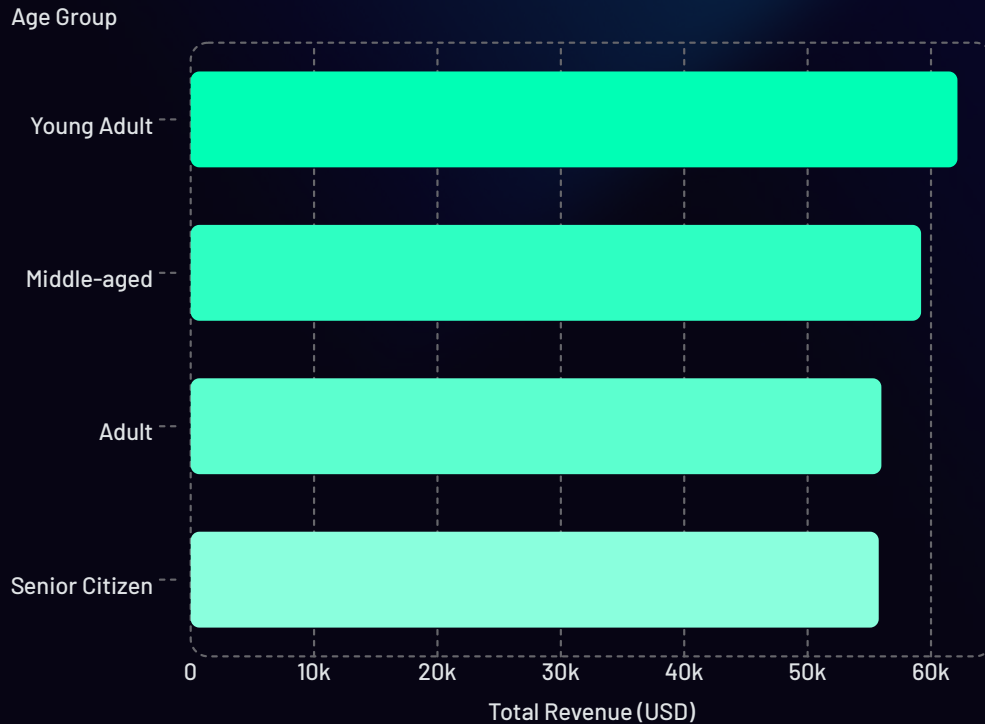
Segments by Previous Purchases

Loyal (3,116) · **Returning** (701) · **New** (83)

Average Purchase Amount by Shipping

Express: **\$60.48** vs Standard: **\$58.46**

Subscription & Age Group Insights



Subscription Status

Subscribers: **1,053** · Non-subscribers: **2,847**

Repeat buyers (5+ purchases): **2,518 non-subscribers** vs 958 subscribers — major untapped opportunity.

📌 Young Adults lead revenue — prime subscription target.

Power BI Dashboard

- 1 3.9K**
Total Customers
- 2 \$59.76**
Avg Purchase Amount
- 3 3.75**
Avg Rating
- 4 27%**
Subscribers
- 5 73 %**
Non-Subscribers

Filters: Subscription Status · Gender · Category · Shipping Type



Business Recommendations



Boost Subscriptions

Better offers to convert 2,518 non-subscribers repeat buyers to subscribers.



Customer Loyalty Programs

Reward Returning customers to grow Loyal segment.



Targeted Marketing

Focus on Young Adults & Express shipping users.



Product Positioning

Highlight top-rated & best-selling items in campaigns.

